KRISH GAUR

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EDUCATION

B.Tech, Computer Science & Engineering | MERI College (affiliated to MDU) | Expected 2029

EXPERIENCE

AI & ML Intern | Coding Arena | Mar 2026 – Jun 2026

- Engineered **RAG pipelines for LLM-powered applications**, integrating embedding generation, vector search, and contextual retrieval to ground model responses in application-specific data.
- Developed **FastAPI-based Python backend services and REST APIs** for LLM inference, NLP pipelines, and ML data-processing workflows, integrating model outputs into application services.
- Evaluated and debugged **LLM outputs** through prompt iteration, test-case validation, response-quality checks, and pipeline configuration tuning to improve reliability across application-specific workflows.

Software Trainee | Duastro | Feb 2025 – May 2025

- Maintained and optimized the company website, implementing **SEO, content, and on-page improvements** to improve search visibility and website performance.

PROJECTS

Jailbreak Oracle: Adaptive LLM Red-Teaming & Vulnerability Discovery Platform | Python, FastAPI, PostgreSQL/pgvector, LiteLLM, Next.js, React - https://github.com/Vagisha13/JailBreak-Oracle

- **Addressed the limitations of static jailbreak testing**, where fixed prompt datasets and manually crafted attacks provide limited coverage against evolving LLM defenses, by engineering an adaptive red-teaming platform that automatically generates and evaluates attack strategies.
- **Designed a multi-agent security architecture** comprising Attacker, Evaluator, Verifier, and Defender agents, with campaign orchestration and BYO-API testing through LiteLLM to test authorized LLM endpoints across providers.
- **Introduced an independent verification layer** that cross-checks evaluator findings before classifying an attack as successful, while pgvector-backed retrieval and evidence-grounded reporting provide traceable vulnerability findings instead of relying solely on model-generated judgments.

ForgeX: AI-Powered Digital Forensics & Investigation Platform | Python 3.12, WebSockets, RAG | NTRO Problem Statement (SIH26148) - https://forge-x-bice.vercel.app/

- **Addressed the challenge of investigating fragmented forensic evidence** across heterogeneous system sources by designing ForgeX as a unified platform for evidence collection, querying, correlation, and reproducible case workflows.
- **Designed FQL (Forensic Query Language)** to provide investigators with a domain-specific interface for querying forensic evidence, supported by OS-level evidence collectors, authenticated case management, and real-time investigation services.
- **Introduced evidence-grounded AI analysis** using embeddings, RAG, and structured outputs to analyze collected evidence while maintaining traceability between generated findings and their underlying evidence.
- **Built and demonstrated the working prototype**, placing **3rd in Code Manthan**, MERI College's internal SIH round.

GhostLine-AI: Enterprise AI Knowledge & Document Intelligence Platform | Python, FastAPI, RAG, FAISS, Embeddings - https://ghost-line-ai-delta.vercel.app/

- **Addressed the difficulty of searching and reasoning over fragmented private documents** such as PDFs, SOPs, contracts, and reports by building an AI knowledge platform that enables users to query proprietary information through natural language.

- **Engineered a document-to-answer RAG pipeline** using document processing, intelligent chunking, embedding generation, and FAISS-based semantic retrieval to identify relevant context before generating responses.
- **Implemented citation-grounded responses and multi-document retrieval**, enabling answers to reference the underlying source content rather than relying solely on the language model's internal knowledge.
- **Extended the system beyond basic document Q&A** with conversational context, streaming responses, and a FastAPI backend designed to support scalable AI knowledge workflows.

ACHIEVEMENTS

- **5× Hackathon Winner**; **Top-5 finisher in 25+ of 50+ hackathons** across AI/ML, Generative AI, cybersecurity, blockchain, and web development.
- **3rd Place — Code Manthan**, MERI College internal hackathon (SIH preparation), for **ForgeX**, an AI-powered digital forensics and investigation platform.

COMMUNITY & EVENTS

- Participated in **20+ technical events, workshops, and developer sessions** hosted by organizations including **Microsoft, Google, Paytm, and Master Union**.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java
AI/ML: LLMs, Generative AI, RAG, NLP, Deep Learning, Embeddings, Vector Databases, Prompt Engineering
Frameworks/APIs: FastAPI, Node.js, React, Next.js, REST APIs, OpenAI, LangChain, LiteLLM, Ollama, Gemini, Groq
Data/Infra: PostgreSQL, pgvector, MongoDB, MySQL, Supabase, Redis, Kafka, NATS, Git